

ETHAN SIMON

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Professional Summary

Junior Mechanical Engineer student seeking a research experience or internship. Prior experience in CAD modeling, simulation and data acquisition. Designed and implemented a flexural fatigue test machine as well as other stress and functionality testing. Design and built multicopter and Tail Sitter drones and integrated with electronic systems as well manufactured parts and created assembly jigs. Additionally, created simulations in MATLAB and Python to inform designs. Work experience developing a GUI for control of a liquid propulsion rocket test stand.

EDUCATION

Case Western Reserve University

Bachelor of Science in Mechanical Engineering

Cleveland, OH

Aug 2024 – May 2028

- **Michalson Morley Scholar**
- **Relevant Coursework:** Dynamics, Thermodynamics, ODE's, PDE's, Design and Manufacturing I/II, Statics

PROFESSIONAL EXPERIENCE

Tylok International

Mechanical Engineering Co-OP

Cleveland, OH

Jan 2026 – Aug 2026

- Full design, manufacturing, and implementation of a flexural fatigue test machine for internal testing in accordance with ASTM standards to test a variety of parts and sizes.
- Developed other internal stress and verification testing apparatus including thermal testing, flow testing, tensile testing, and pressure testing.
- Worked with the sales team to improve product showcasing and system demonstrations for trade shows.
- Researched, acquired and integrated data acquisition with a Dewesoft system, developed and used programs in DewesoftX as well as C++ for technical analysis.

Camp Ben Frankel

Camp Counselor

Carbondale, IL

Jun 2025 – Aug 2025

- Mentored and supported children ages 8-13
- Planned and led activities - rocketry, archery, board games, swimming etc.
- Facilitated team building to increase camaraderie, and built trust and rapport

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

VTOL CWRU

Mechanical and Aerospace Team Lead

Aug 2024 – Present

- Worked on mechanical design of frames and payload mechanisms
- Design of Tail Sitter for heavy lift application, Hexacopter, Quadcopters, and Tiltrotors
- Used SolidWorks for CAD and Simulation
- Developed project planning documents, safety documents, and work instructions
- Developed scripts in MATLAB and Python for simulation and to inform design decisions

Liquid Propulsion Team

Aug 25 – Jan 26

- Developed a control GUI in PyQt5 that integrate the mechanical control with the software to run hot fire sequences
- Designed mechanical control systems to regulate thrust using solenoids and servos

University of Michigan Battery Lab

2022-2024

- Physical analysis and data collection of single and multi-cell batteries organized in spread sheets.
- Python based analysis of cells using libraries including Pandas, Scikit-learn, SciPy, and hdbscan

Sigma Chi Fraternity

- Scholarship Chair: organized mentorship programs, connected members with resources, facilitated career workshops
- House Manager: coordinated housing of the chapter, helped develop housing corps, communicated with CWRU and other orgs.

SKILLS & INTERESTS

- **Skills:** SolidWorks, MATLAB, Python, Arduino, Microsoft Office Suite, QGroundControl, Machining, Mechanical Analysis
- **Interests:** Soccer, Fishing, Hiking, Cooking, Woodworking, Catan, Weight-Lifting, Music,